

Evidence Summary - Technology #13

Self-Referencing Piezoresistive Pressure Sensor

External Evidence Sources (5 total):

Source 1: Chronically Implanted Pressure Sensors: Challenges and State of

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC4279503>

Excerpt: Drift and Long-Term Accuracy. To achieve reliable long-term measurements, implanted sensors should possess a stable, consistent response over their lifetime.

Source 2: A simple and effective method to compensate the thermal drift of

URL: <https://www.sciencedirect.com/science/article/abs/pii/S0924424724006095>

Excerpt: This article presents a study of implantable blood pressure sensors based on a probe made up of four piezoresistors. An original method combining both ...

Source 3: Compensation of Pressure Sensor Drifts - Encyclopedia.pub

URL: <https://encyclopedia.pub/entry/52473>

Excerpt: Dec 8, 2023 · To compensate induced drift in measurements, several neural networks were devised and be applied to stresses caused by two thermal cycles: 260 C ...

Source 4: Implantable Blood Pressure Sensors with Analogic Thermal Drift - MDPI

URL: <https://www.mdpi.com/2673-4591/6/1/34>

Excerpt: May 17, 2021 · To quantify and compensate for the thermal drift, we developed the equivalent model of the piezoresistive probe by using the Cadence software.

Source 5: [PDF] piezoresistive pressure sensor with dual-unit configuration for

URL: https://engineering.purdue.edu/oxidemems/conferences/transducers2013/PDFs/Papers/444_0996.pdf

Excerpt: For the first time, we propose an on-chip dual-unit sensor configuration in this study to eliminate temperature-drift and packaging-stress of pressure sensors.

*Evidence Classification: This dossier contains source-native information, derived engineering analysis, engineering proposals, and unresolved questions.
No evidence has been physically validated.*